

2023



Scaling and Enhancing Security for Electric Vehicle Chargers Using AWS IoT Core with Wallbox

Learn how electric vehicle charging company Wallbox migrated its connectivity infrastructure to AWS IoT Core.

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50% reduction

in total cost of operability per charger

Achieved automatic scalability

for millions of devices



Enhanced

compliance and security

Increased speed and reliability

of network connections

Overview

[Wallbox](#), a leading provider of smart chargers for electric vehicles, was looking to scale its Internet of Things (IoT) infrastructure to service its more than 500,000 smart devices globally. The company aggregated metrics and analytics from those smart devices in a data lake for customer service and service engineering purposes and for visibility into the key performance indicators of its devices. To get those necessary insights, Wallbox needed a solution that could provide the scalability to collect near-real-time events from its devices.

Having used Amazon Web Services (AWS) since its inception, Wallbox turned to [AWS IoT Core](#), which helps businesses connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure. By migrating its connectivity infrastructure to AWS, Wallbox reduced its total cost of operability per charger by 50 percent, increased scalability, enhanced compliance and security, and increased the speed and reliability of its smart charger network connections.



Opportunity | Using AWS IoT Core to Increase Scalability and Deliver Insights

Wallbox is a Spanish producer of electric vehicle charging and energy management solutions that serves more than 100 countries. Its goal is to give customers the power to create and manage their energy consumption, save on electricity bills, and live more sustainably.

Wallbox began as a cloud-native company using AWS in 2015. However, it wanted to improve its architecture so that it could better scale to support its more than 500,000 charging devices worldwide. The company needed a solution that was compatible with the industry-standard MQTT communication protocol for charging devices. “The main driver for migration was scalability,” says Darío Fernández Barrio, Director of IT, Cloud and Security at Wallbox. “By using AWS IoT Core, we could get near-real-time events at scale for analytical purposes to deliver value to the business.”

The company performed its migration in two phases. First, Wallbox migrated away from its legacy database architecture, which was used for extracting data and insights from the Wallbox chargers for quick actioning and advanced analytics. Next, Wallbox migrated to managed MQTT from AWS IoT Core for its connectivity solution to send and receive information to and from smart chargers. The insights gleaned from the transmitted data help end users make energy-saving choices and give the company the ability to provide live support and configurations to the devices. The original model would disconnect after 2 hours of connectivity, but the new solution is more reliable because of the high uptime provided by AWS IoT Core.

Although Wallbox has hundreds of thousands of devices in the field, mostly in domestic settings, the first part of the migration focused only on its 1,000 public chargers. Now that Wallbox is fully using AWS IoT Core, which scales automatically, the company is migrating the rest of its 500,000 devices without worrying about managing infrastructure.

Solution | Creating a Seamless Architecture and Enhancing Security Using AWS IoT Core

Wallbox’s new connectivity architecture, centered on AWS IoT Core, interacts with other AWS services to create a seamlessly connected system that can transmit data in near real time for actionable insights. Instead of querying



databases, the company now uses an event-driven architecture. AWS IoT Core communicates with the fully managed [Amazon Kinesis](#), an event streaming system that cost-effectively processes and analyzes streaming data at any scale. For data ingestion, the company also uses [AWS Lambda](#), a serverless, event-driven compute service. “With the ability to use AWS IoT Core seamlessly with other services, we can iterate very quickly and prototype with ease,” says Fernández Barrio. “Everything is connected and tightly integrated.”

Using AWS IoT Core takes the guesswork out of compliance and regulatory requirements for Wallbox. Because AWS IoT Core comes with robust certification and connectivity security features like mutual authentication and end-to-end encryption, Wallbox can deploy certificates for each charging device in a quick, efficient manner. Every device uses AWS IoT Core, so the company can quickly assign policies, create groups, or revoke certificates when identifying security problems. Similarly, using [AWS IoT Device Management](#)—a service to register, organize, monitor, and remotely manage connected devices at scale—Wallbox can group devices and perform targeted actions, such as performing software updates.

Wallbox has also used device data to create an antitampering system that recognizes when a device has been manipulated in the field. “By using AWS IoT Core, we reduce a lot of engineering hours otherwise needed to maintain our security posture,” says Fernández Barrio. “Our devices are already compliant with the many electric vehicle charging regulations that exist in the countries we operate in.”

Outcome | Exploring Artificial Intelligence Solutions to Improve Customer Support

Wallbox continues to look for innovative ways to improve the capabilities of its smart chargers. With its new connectivity solution using AWS IoT Core, the company can help its customers optimize energy management. Its data analytics teams are venturing into possible artificial intelligence solutions to improve customer support using the data collected from the devices. Wallbox also plans to provide its support teams with access to near-real-time analyses of charger performance to streamline troubleshooting.

“The key for us is to provide a good experience for the end user,” says Fernández Barrio. “It’s essential to create a secure and reliable method of communication with the devices.”

About Wallbox

Wallbox is a Spain-based producer of electric vehicle smart-charging systems and energy management solutions that combine innovative technology with outstanding design to manage the communication between vehicles, grids, buildings, and chargers.

AWS Services Used

AWS IoT Core



AWS IoT Core lets you connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure.

[Learn more »](#)

AWS IoT Device Management

AWS IoT Device Management helps you register, organize, monitor, and remotely manage IoT devices at scale.

[Learn more »](#)

AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

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Amazon Kinesis

Amazon Kinesis cost-effectively processes and analyzes streaming data at any scale as a fully managed service.

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